The solution for all adverse environmental impacts lies in handling fossil fuels and materials better.

CO_2 the Largest Culprit

The UN system and international treaties identify seven gases as GHGs, as included in Annex A to the Kyoto Protocol. These are CO_2; methane (CH_4); nitrous oxide (N_2O); hydrofluorocarbons (HFCs); perfluorocarbons (PFCs); sulphur hexafluoride (SF_6); and nitrogen trifluoride (NF_3). The last gas, NF_3, was added by the Doha Amendment.

All parties to the UN Framework Convention on Climate Change, 1994 (the Climate Change Convention), have committed to publish and make available to the Conference of Parties, 'national inventories' of anthropogenic emissions (largely the result of industrial production and urbanized living) 'by sources and removals by sinks of all greenhouse gases' except the 'gases controlled by the Montreal Protocol'.

There are different reporting obligations for the industrialized countries that are further divided in three broad categories: first, the countries responsible for much of the GHG emissions until 1990; second, industrialized countries that are not considered Economies in Transition (EIT); and third, the remaining industrialized member countries. There are, in all, forty-three such countries, and are collectively known as Annex I countries. These include the EITs, which are also separately named in Annex II of the Convention. The Convention has adopted comprehensive guidelines on reporting requirements and processes.

CO_2 is the largest GHG emitted every year. In 2018, total GHG emissions from anthropogenic sources were 51.8 gigatons of CO_2 ($GtCO_2$) equivalent. Including land-use related emissions, total GHG emissions were 55.6 $GtCO_2$. CO_2 made up 72 per cent of global GHGs, followed by methane (CH4) at about 19 per cent and nitrous oxide (N_2O) at about 6 per